

Notes: Thompson (JFE, 2011)
Simple formulas for standard errors that cluster by both firm and time

Contents

1	Big picture	2
2	Data, objects, and dependence structures	2
2.1	Panel regression object	2
2.2	Firm effects, time effects, and persistent common shocks	2
2.3	Empirical application data	2
3	Models and standard-error formulas	3
3.1	Two-way clustering without persistent common shocks	3
3.2	Two-way clustering with persistent common shocks	3
3.3	Bias-variance trade-off	3
3.4	Monte Carlo design	3
3.5	Industry profitability application	3
4	Identification and empirical design logic	3
5	Section-by-section study guide	3
6	Tables and figures	4
6.1	Table 1: Monte Carlo comparison of standard error formulas – Panel A, iid benchmark	4
6.2	Table 1: Monte Carlo comparison of standard error formulas – Panel B, firm and time effects	4
6.3	Table 1: Monte Carlo comparison of standard error formulas – Panel C, persistent common shocks	5
6.4	Table 2: Application to modeling industry profitability	5
7	Short summary	6
8	Conclusion	6
8.1	Core conclusion	6
8.2	Economic intuition	7
8.3	Practical implications	7
8.4	Limitations	7
8.5	Takeaway	7

1 Big picture

- **Question.** How should researchers compute standard errors in finance panel regressions when residuals may be correlated both within firms over time and across firms within the same time period?
- **Core contribution.** The paper gives a simple two-way cluster variance formula: cluster by firm, add cluster by time, and subtract the White heteroskedasticity-robust matrix to avoid double-counting.
- **Main formula.** In the simplest case, the covariance estimator is $\hat{V}_{firm} + \hat{V}_{time,0} - \hat{V}_{white,0}$.
- **Persistent common shocks.** The paper extends the formula to allow shocks that affect many firms and persist across time.
- **Monte Carlo lesson.** Pure double clustering works well when both dimensions have around 25 or more observations; persistent common-shock corrections require a longer time dimension.
- **Economic takeaway.** Statistical significance in finance panels is a joint statement about coefficients and covariance structure.

2 Data, objects, and dependence structures

2.1 Panel regression object

The basic object is

$$y_{it} = x'_{it}\beta + e_{it}.$$

The language of the paper uses firms and time, but the method generalizes to any two clustering dimensions.

Interpretation. The key input is a panel structure with two dimensions along which residuals may be dependent.

2.2 Firm effects, time effects, and persistent common shocks

- Firm effects allow arbitrary serial correlation within a firm.
- Time effects allow arbitrary cross-sectional correlation across firms at a date.
- Persistent common shocks allow correlation between different firms in different periods up to lag L .

The paper's intuition is summarized by

$$e_{it} = h'_i f_t + Z_{it} + u_{it}, \quad Z_{it} = \phi Z_{i,t-1} + B_{it}.$$

Interpretation. Z_{it} generates firm effects; $h'_i f_t$ generates time effects and, if f_t is persistent, persistent common shocks.

2.3 Empirical application data

The application uses annual Compustat data from 1964–2007. Industries are four-digit SIC categories. The sample contains 434 categories, 44 years, and 15,066 industry-year observations.

Interpretation. The application is chosen because both clustering dimensions matter: industry and year.

3 Models and standard-error formulas

3.1 Two-way clustering without persistent common shocks

$$\widehat{Var}(\hat{\beta}) = \widehat{V}_{firm} + \widehat{V}_{time,0} - \widehat{V}_{white,0}.$$

Interpretation. The subtraction removes double-counting of the diagonal observation-level products.

3.2 Two-way clustering with persistent common shocks

$$\widehat{Var}(\hat{\beta}) = \widehat{V}_{firm} + \widehat{V}_{time,0} - \widehat{V}_{white,0} + \sum_{l=1}^L (\widehat{V}_{time,l} + \widehat{V}'_{time,l} - \widehat{V}_{white,l} - \widehat{V}'_{white,l}).$$

Interpretation. The lag terms capture correlation between time-cluster sums at different dates.

3.3 Bias–variance trade-off

More robust covariance estimators have less omitted-dependence bias but more estimation variance. In small samples, noisy covariance estimates can cause false rejections.

Interpretation. The most robust formula is not automatically best in every finite sample.

3.4 Monte Carlo design

$$y_{it} = \beta_0 + \beta_1 x_{1,it} + \beta_2 x_{2,it} + e_{it}, \quad \beta_0 = 0, \quad \beta_1 = \beta_2 = 1.$$

The paper simulates 5,000 samples and reports rejection frequencies for true nulls. A correct 5% test should reject close to 5% of the time.

3.5 Industry profitability application

$$ROA_{m,t} = \beta_0 + \beta_1 \ln(HHI_{m,t-1}) + \beta_2 PB_{m,t-1} + \beta_3 DB_{m,t-1} + \beta_4 ROA_{t-1} + e_{m,t}.$$

Interpretation. The application shows that the right clustering dimension differs by regressor.

4 Identification and empirical design logic

This is not a causal identification paper. The key object is the true sampling variance of OLS coefficients under two-dimensional dependence. One-way clustering can fail because clustering by firm ignores time shocks and clustering by time ignores persistent firm-specific shocks. Double clustering works by an inclusion–exclusion logic.

5 Section-by-section study guide

- **Introduction:** motivates two-way dependence and states the simple formula.
- **Section 2:** defines firm effects, time effects, and persistent common shocks.

- **Section 3:** derives the covariance formulas.
- **Section 4:** discusses when robust standard errors help and when they create small-sample distortions.
- **Section 5:** uses Monte Carlo simulations to assess test size.
- **Section 6:** applies the method to industry profitability.

6 Tables and figures

The paper contains two main tables and no figures. The tables are directly cropped from the paper. Table 1 is split into three panels for readability.

6.1 Table 1: Monte Carlo comparison of standard error formulas – Panel A, iid benchmark

Read: Panel A is the placebo benchmark. OLS and simple clustering are close to the correct 5% size, while persistent-common-shock corrections over-reject in small samples.

Detailed interpretation. Panel A shows the cost of using an overly robust estimator when there is no dependence problem to solve.

Economic message. Robust inference is not free: unnecessary autocovariance corrections can create false significance in small samples.

<i>Panel A: Both regressors and ε_{it} are iid $N(0,1)$ across both i and t, so OLS error assumptions are satisfied</i>						
	<i>T=25, N=50</i>		<i>T=50, N=50</i>		<i>T=100, N=100</i>	
	β_1	β_2	β_1	β_2	β_1	β_2
OLS std errors	0.049	0.048	0.050	0.047	0.049	0.056
Cluster by firm	0.053	0.056	0.056	0.053	0.052	0.055
Cluster by time	0.058	0.065	0.057	0.056	0.052	0.059
Cluster by firm, time FE	0.057	0.056	0.060	0.058	0.051	0.058
Cluster by time, firm FE	0.065	0.067	0.060	0.059	0.051	0.062
Cluster by both firm and time (not robust to persistent common shocks)	0.069	0.070	0.062	0.064	0.054	0.059
Cluster by both firm and time (robust to persistent common shocks, $L=2$)	0.127	0.123	0.100	0.100	0.069	0.078

Table 1 Panel A: iid errors and regressors

6.2 Table 1: Monte Carlo comparison of standard error formulas – Panel B, firm and time effects

Read: Panel B introduces firm and time effects but no persistent common shocks. OLS over-rejects badly. Single clustering works only for one coefficient at a time. Double clustering performs well as sample size grows.

Detailed interpretation. This panel shows that one covariance choice may be right for one regressor and wrong for another. Double clustering handles both sources simultaneously.

Economic message. When different regressors load on different dependence dimensions, single clustering can create selective inference errors.

Panel B: Errors have both time and firm effects, and persistent common shocks. x_{1it} has time fixed effects (and no persistent common shock), and x_{2it} has firm effects that follow an autoregressive process

	T=25, N=50		T=50, N=50		T=100, N=100	
	β_1	β_2	β_1	β_2	β_1	β_2
OLS std errors	0.560	0.425	0.519	0.482	0.640	0.478
Cluster by firm	0.695	0.058	0.635	0.059	0.707	0.053
Cluster by time	0.093	0.531	0.074	0.533	0.056	0.501
Cluster by firm, time FE	–	0.058	–	0.058	–	0.054
Cluster by time, firm FE	0.093	0.369	0.074	0.466	0.056	0.477
Cluster by both firm and time (not robust to persistent common shocks)	0.105	0.066	0.081	0.061	0.060	0.055
Cluster by both firm and time (robust to persistent common shocks, L=2)	0.174	0.103	0.113	0.080	0.076	0.062

Panel C: Errors have both time and firm effects, and persistent common shocks. x_{1it} has time effects with persistent common shocks, and x_{2it} has firm fixed effects

Table 1 Panel B: firm and time effects

6.3 Table 1: Monte Carlo comparison of standard error formulas – Panel C, persistent common shocks

Read: Panel C adds persistent common shocks. Simple double clustering remains too liberal for the time-varying coefficient, while the persistent-shock correction improves inference but still over-rejects in smaller samples.

Detailed interpretation. Persistent common shocks create cross-firm and cross-time dependence, which pure two-way clustering does not fully capture.

Economic message. Persistent aggregate shocks require additional corrections, but applied researchers need a sufficiently long time dimension.

Panel C: Errors have both time and firm effects, and persistent common shocks. x_{1it} has time effects with persistent common shocks, and x_{2it} has firm fixed effects

	T=25, N=50		T=50, N=50		T=100, N=100	
	β_1	β_2	β_1	β_2	β_1	β_2
OLS std errors	0.747	0.563	0.765	0.673	0.809	0.750
Cluster by firm	0.906	0.074	0.916	0.074	0.931	0.056
Cluster by time	0.169	0.865	0.155	0.906	0.139	0.924
Cluster by firm, time FE	0.654	0.074	0.672	0.074	0.726	0.056
Cluster by time, firm FE	0.167	–	0.157	–	0.140	–
Cluster by both firm and time (not robust to persistent common shocks)	0.176	0.087	0.162	0.081	0.143	0.058
Cluster by both firm and time (robust to persistent common shocks, L=2)	0.201	0.127	0.145	0.100	0.098	0.067

Table 1 Panel C: persistent common shocks

6.4 Table 2: Application to modeling industry profitability

Read: The estimate on $\ln(HHI)$ is positive, 0.0049. White standard errors give a t-statistic of 10.53, while industry and double-clustered t-statistics are around 4.3–4.6. Price-to-book, dividends-to-book, and market ROA are also positive predictors.

Detailed interpretation. The table shows that each regressor’s dependence pattern matters. Industry concentration varies mostly across industries; market ROA varies over time. Double clustering gives one coherent inference approach for both.

Economic message. Empirical conclusions can survive robust inference, but t-statistics often fall sharply once the covariance structure matches the panel.

company management anticipates higher earnings. $\overline{ROA}_t$ is the return on assets for the market as a whole in year t . It controls for market-wide trends in profitability. The data and regressors are described in more detail in Appendix B.

Notice that this example uses industry-level rather than firm-level data. Even though the text of this paper has mostly referred to the clustering dimensions as firms and time periods, the results trivially generalize to clustering along any two dimensions. I picked this example in part because it is

and the smallest effect is for $\ln(HHI)$. This makes sense, since $\overline{ROA}$ is positively correlated across time, while $\ln(HHI)$ has very weak time effects.

7. Conclusion

This paper derives easy-to-compute formulas for standard errors that cluster by both firm and time. Both the statistical theory and the Monte Carlo results suggest that

Table 2

Application to modeling industry profitability.

The table shows a regression to model industry profitability. The dependent variable is $ROA_{m,t}$, the ratio of earnings-to-assets in industry m in year t . The regressors are as follows. $\ln(HHI_{m,t-1})$ is the log of the Hefindahl-Hirschman concentration index for the industry, computed from market caps. Price/Book equity $_{m,t-1}$ is the price-to-book ratio. Dividends/Book equity $_{m,t-1}$ is the dividends-to-book ratio. Market ROA_{t-1} is the market-wide (not industry-specific) ratio of earnings-to-assets. “Estimate” denotes ordinary-least-squares estimates. t -Statistics are presented for different standard error formulas: (1) “White” denotes V_{white} , (2) “single-clustered, time” denotes V_{time} , (3) “single-clustered, firm” denotes V_{firm} , (3) “double-clustered, $L=0$ ” denotes $V_{time} + V_{firm} - V_{white}$, and (4) “double-clustered, $L=2$ ” denotes $V_{firm} + V_{time} - V_{white} + \{\text{corrections for persistent common shocks}\}$ with $L=2$. See Appendix B for details about the data.

Regressor	Estimate	White	t -Statistics			
			Single-clustered		Double-clustered	
			Time	Industry	$L=0$	$L=2$
$\ln(HHI_{m,t-1})$	0.0049	10.530	9.988	4.314	4.274	4.610
Price/Book equity $_{m,t-1}$	0.0072	17.670	5.604	11.062	5.212	3.394
Dividends/Book equity $_{m,t-1}$	0.3171	20.700	9.877	10.083	7.505	5.482
Market ROA_{t-1}	1.0631	32.720	8.667	19.958	8.195	4.924
Intercept	−0.0521	−14.461	−9.148	−5.847	−5.240	−4.714
R -squared: 19.29%						

Table 2: application to modeling industry profitability

7 Short summary

- The basic two-way clustered covariance estimator is $V_{firm} + V_{time} - V_{white}$.
- The formula is an inclusion–exclusion rule for covariance products.
- Persistent common shocks require additional lagged autocovariance terms.
- More robust formulas reduce bias but can increase small-sample over-rejection.
- Monte Carlo results support simple double clustering in panels with roughly 25 observations in both dimensions.
- The empirical application shows that clustering choices materially alter t-statistics.

8 Conclusion

8.1 Core conclusion

Two-way clustered standard errors are conceptually necessary and practically easy in many finance panel regressions.

8.2 Economic intuition

Firm clustering collects within-firm covariance terms; time clustering collects within-time covariance terms; the White matrix is counted twice and must be subtracted.

8.3 Practical implications

Applied researchers should match standard errors to both error dependence and regressor dependence. A single cluster dimension may be right for one coefficient but wrong for another.

8.4 Limitations

Persistent-common-shock corrections have poor small-sample behavior and require long time dimensions.

8.5 Takeaway

Correct inference in finance panels requires modeling dependence. Thompson provides the simplest practical formula for the common case of two-way panel dependence.